Matrices and Determinants

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Part I

Matrices Problem Set

Chapter 1

Matrices

Q: Invent a 3×3 magic matrix M with the entries 1 , 2 , 3 , ...9. All rows and columns and diagonals add to 15 . The first row could be (8, 3, 4) . What could be M.

Q: If $A = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$. Find $A^T A$.

Q: Find the adjoints of the following matrices

a) $\begin{bmatrix} 15 & 7 \\ 8 & 14 \end{bmatrix}$

b) $\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Q: If $A = \begin{bmatrix} 2x+6 \\ 26 \\ 2z+48 \\ t^2+41 \end{bmatrix}$

$$= \begin{bmatrix} x^2 & -x & 1 \\ y^2 & -y & 2 \\ z^2 & -z & 3 \\ t^2 & -t & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Then , A can be

a) $\begin{bmatrix} 10 \\ 26 \\ 46 \\ 47 \end{bmatrix}$

b) $\begin{bmatrix} 10 \\ 26 \\ 54 \\ 42 \end{bmatrix}$

c) $\begin{bmatrix} 10 \\ 26 \\ 46 \\ 42 \end{bmatrix}$

d) $\begin{bmatrix} 10 \\ 26 \\ 54 \\ 41\frac{9}{25} \end{bmatrix}$

1.0.0.1 Linked Comprehension Type Questions

Comprehension 1: Two vectors $v = (1, 2, 3)$ and $w = (1, 3, 4)$ are taken and put in the columns of a matrix as

$$M = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$$

The linear combination of these two vectors can be expressed as

$$c \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + d \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix}$$

Q1: A relation between A , B , c and d is

a) $6c + 8d = A + B$
b) $8c + 6d = A + B$
c) $6c + 8d = 6A + 8B$
d) None of these

Q2: The matrix $\begin{bmatrix} A \\ B \end{bmatrix}$ in the equation

$\begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix}$ is given by

a) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
b) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
c) $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$
d) None of these

1.0.0.2 Matrix Match Type Problems

Matrix 1 : Three elementary matrices $E = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$, $G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ are given. Under column I , resultant matrices (lower triangular) obtained from E, F, G or their inverses are given. Match the matrices given in column I with their corresponding expressions given under column II :

Column I	Column II
(P) $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$	(A)EFG
(Q) $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}$	(B)$EF^{-1}G$
(R) $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$	(C)$E^{-1}FG^{-1}$
(S) $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$	(D)$E^{-1}F^{-1}G^{-1}$
	(E)$G^{-1}F^{-1}E^{-1}$

Chapter 2

JEE Mains Problems

Q1 : Let $A = \begin{bmatrix} 1 & 2 \\ 1 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 \\ \beta & 2 \end{bmatrix}$. If $A^2 - 4A + I = 0$ and $B^2 - 5B - 6I = 0$, then among the two statements :

(S1) : $[(B - A)(B + A)]^T = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$

(S2) : $det\,(adj\,(A + B)) = -5$,

[JEE Mains 2nd Apr 2026 Shift 1 Q4]

(A) Only (S1) is correct

(B) Only (S2) is correct

(C) Both (S1) and (S2) are correct

(D) Both (S1) and (S2) are wrong

{Solution : https://www.youtube.com/watch?v=$_uTUbsovqU8$}